# COLIMITS OF SETS AND MAPS OUT OF A FINITE SET

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for both injectivity and surjectivity, the finiteness of the set being used explicitly, and no step is left to the reader.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The comparison map of an inductive system. Same chapter, the displayed comparison map of an inductive system used in the smallness discussion. Supplies the map whose bijectivity is the assertion.
- Colimits of sets and their universal property. Sets, the description of the colimit of a system of sets by equivalence classes of elements. Supplies that two elements of two different stages become equal in the colimit exactly when they become equal in a later stage, which is the mechanism of the proof.

*Source reference.* Original equation 05NN: sources/stacks-injectives/pinned-original-source.tex, source lines 279–279. Navigation pointer invoked at proof line 287; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Suppose that, in (05NN), $\mathcal{C}$ is the category of sets and $A$ is a finite set, then the map is a bijection.*

*Proof.* Let $f : A \to \operatorname{colim} B_\beta$. The range of $f$ is finite, containing say elements $c_1, \ldots, c_r \in \operatorname{colim} B_\beta$. These all come from some elements in $B_\beta$ for $\beta \in \alpha$ large by definition of the colimit. Thus we can define $\widetilde{f} : A \to B_\beta$ lifting $f$ at a finite stage. This proves that (05NN) is surjective. Next, suppose two maps $f : A \to$

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$B_\gamma, f' : A \to B_{\gamma'}$ define the same map $A \to \operatorname{colim} B_\beta$. Then each of the finitely many elements of $A$ gets sent to the same point in the colimit. By definition of the colimit for sets, there is $\beta \geq \gamma, \gamma'$ such that the finitely many elements of $A$ get sent to the same points in $B_\beta$ under $f$ and $f'$. This proves that (05NN) is injective. $\square$